IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

ANALYSIS AND DESIGN OF CIRCUITS

Friday 5 May 14:30

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions. Question One carries 40% of the marks. Question 2 and 3 carry equal marks (30% each)

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A.S. Holmes, P.D. Mitcheson
Second Marker(s) :

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Information for Candidates

Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$ while the phasor voltage is denoted by X.

- Component and source values are generally given in Ohms, Farads, Henrys, Volts or Amps with the unit symbol omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.

- Times are in seconds unless otherwise stated.

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The Questions

1. a) For the circuit shown in Figure 1.a, calculate the voltage at node *X*.

[6]

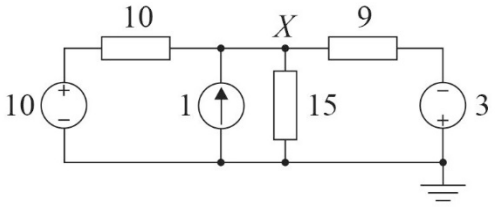


Figure 1.a

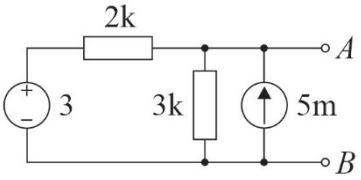


Figure 1.b

b) Draw the Thévenin equivalent of the network shown in Figure 1.b, assigning values to the elements.

[5]

c) For the circuit shown in Figure 1.c, give an expression for the voltage *Z* in terms of the voltages U_1 and U_2.

[6]

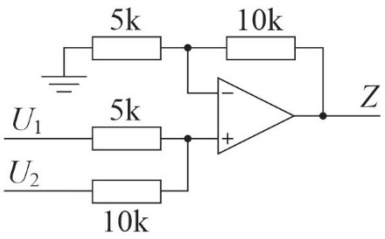


Figure 1.c

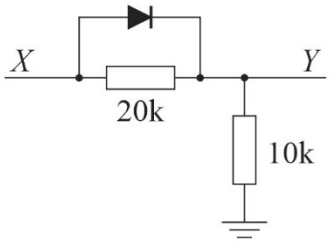


Figure 1.d

d) For the circuit shown in Figure 1.d, draw a dimensioned sketch showing the variation of voltage *Y* with voltage *X*, covering the range $-3 \le X \le 3$. You should assume that the diode has a forward voltage of 0.7 V but is otherwise ideal.

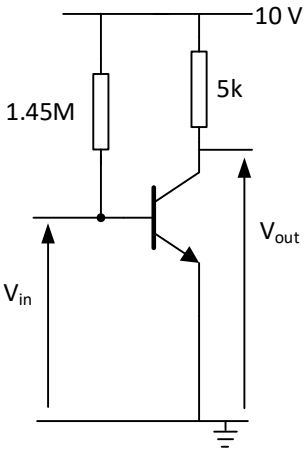
[6]

Question 1 continues on the next page

Question 1 continued

e) The circuit in Figure 1.e is biased such that 1 mA of collector current is flowing. Calculate the small signal input resistance, output resistance and gain of the circuit.

[5]



V_A=180 V

β=150

Figure 1.e

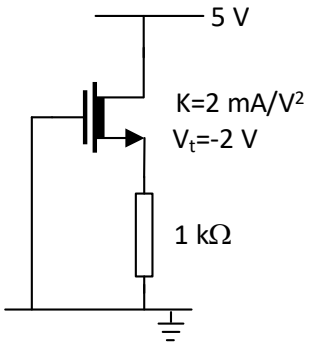


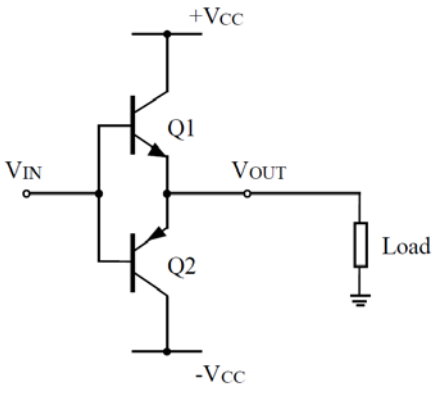
Figure 1.f

f) Calculate the drain current in the MOSFET of Figure 1.f.

[6]

g) The circuits in Figure 1.g are unity voltage gain amplifiers. By sketching waveforms, compare and contrast their performance. What function do resistors R3 and R4 perform?

[6]



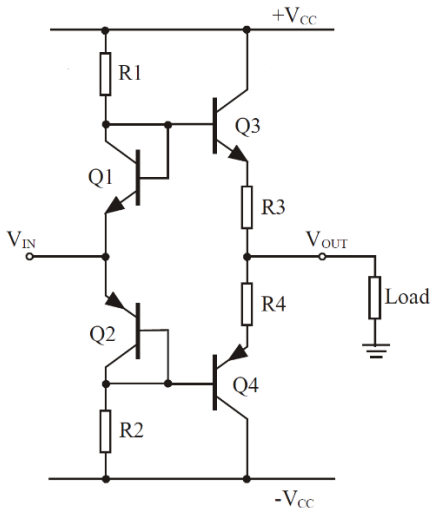


Figure 1.g

2. Figure 2.1 shows an active band-pass filter comprising an opamp in inverting amplifier configuration with an LCR network in the feedback path.

a) Explain briefly, with reference to the individual circuit elements, why the circuit acts as a band-pass filter.

[4]

b) Obtain an expression for the transfer function $H(j\omega) = \frac{Y}{X}$ of the filter, and hence show that the centre frequency and quality factor are given by:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad ; \quad Q = \frac{R_2}{\omega_0 L}$$

[8]

c) If the filter is required to have a centre frequency of 10 kHz, a 3 dB bandwidth of 500 Hz a peak gain of 0 dB, and an input impedance of 1 kΩ, what values should be used for the four passive components?

[8]

d) Assuming the component values you calculated in part c), draw a dimensioned sketch showing the variation with frequency of $|H|$ together with the low- and high-frequency asymptotes.

[6]

e) It is suggested that the same frequency response can be achieved with the filter shown in Figure 2.2. Is this true and, if so, what are the potential advantages of the active circuit over the passive one?

[4]

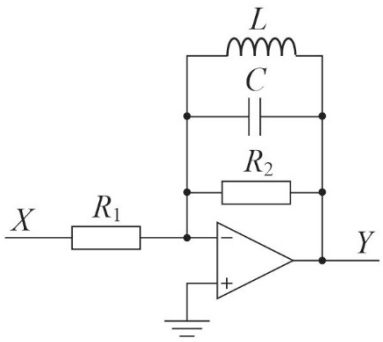


Figure 2.1

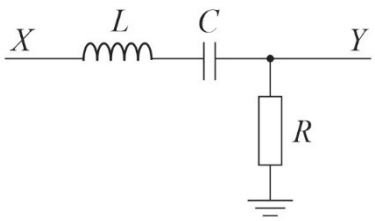


Figure 2.2

3. Figure 3.1 shows a differential amplifier. All transistors are identical.

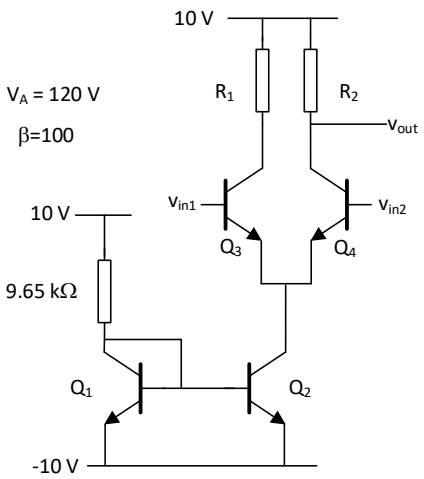


Figure 3.1

a) Calculate values of R_1 and R_2 to bias V_{out} at 0 V with zero differential input.

[2]

b) Calculate the quiescent power consumption of the circuit.

[3]

c) Draw a small signal model of the circuit in differential mode and calculate the single-ended differential gain.

[5]

d) State and explain the common mode voltage range.

[4]

Question 3 continues on the next page

Question 3 continued

It is now proposed to replace R_1 and R_2 with current mirror outputs, as shown in Figure 3.2.

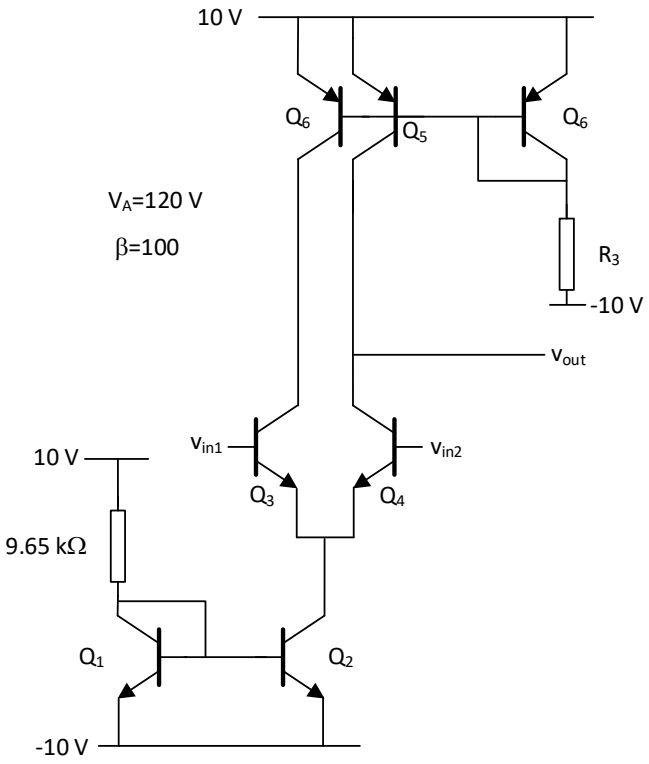


Figure 3.2

e) Calculate the value of R_3 to make the collector current of Q_3 and Q_4 the same as it was in Figure 3.1 and calculate the quiescent power consumption of the circuit of Figure 3.2.
[4]

f) Calculate the common-mode voltage range and then, by drawing a new small signal model, calculate the differential mode gain.
[8]

g) Explain what happens to the output voltage if, due to component tolerances, the output of the current mirror that has replaced R_1 and R_2 is higher or lower than half of the tail current. Assume $v_{in1}=v_{in2}=0$ V.
[4]

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